InsightWear AI – Customer Intent Intelligence Platform

Team: VibeCoders
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Team Leader :Eswar M
Reg.NO:127003059

CATEGORY 4: Business & E-Commerce

Problem 2: Understanding customer intent

1. Problem and Market Analysis

Fashion e-commerce platforms receive thousands of customer reviews daily. Most platforms rely only on star ratings, which hide the real causes behind customer satisfaction or dissatisfaction such as sizing issues, stitching problems, fabric quality concerns, or comfort complaints.

This lack of structured insight causes brands to miss product defects early, leading to higher return rates and delayed product improvement cycles.

Target Customers

• Fashion brands selling on Amazon, Flipkart, and Myntra

• Product design and quality assurance teams

• Marketplace analytics teams

Market Opportunity

The global fashion e-commerce market exceeds $800B. Even mid-size brands process tens of thousands of reviews monthly. Automated review intelligence platforms can significantly reduce product returns and improve product design cycles.

Limitations of Current Solutions

Existing tools such as Brandwatch, Mention, and generic sentiment analysis platforms provide only overall sentiment scores. They fail to extract product-specific insights such as fit accuracy, comfort level, fabric durability, or style perception.

2. Proposed Solution Overview

InsightWear AI is a review intelligence platform that extracts detailed product insights from customer feedback using Natural Language Processing (NLP). The system identifies aspects such as Fit, Comfort, Fabric Quality, and Style and generates sentiment scores for each dimension.

Key Features

• Automated aspect extraction from customer reviews

• Sentiment scoring per product dimension

• Product insight dashboards for design and QA teams

• Early issue detection for product teams

Technical Approach

The system uses lightweight transformer models such as DistilBERT or MiniLM combined with rule-based aspect extraction pipelines. Backend services are implemented using Python APIs and scalable microservice architecture.

Dataset Used for Prototype

The prototype uses the Women's E-Commerce Clothing Reviews dataset (~23,000 reviews) to train and evaluate aspect-based sentiment extraction.

Unique Value Proposition

Unlike generic sentiment analysis tools, InsightWear AI provides fashion-specific aspect intelligence (fit, comfort, fabric quality, and style) enabling brands to identify product design issues early and reduce return rates.

3. Cost Benefit Analysis

Infrastructure Costs (Monthly Estimate)

• Cloud VM (4 vCPU compute server): ₹1500

• Database storage: ₹500

• Bandwidth and misc processing: ₹300

• Monitoring and logging: ₹200

Total estimated infrastructure cost ≈ ₹2500 per month

Maintenance and Support Costs

Developer maintenance, monitoring, and support services are estimated at approximately ₹5000 per month.

Total Operating Cost

Total estimated operational cost ≈ ₹7500 per month

Per Review Cost

If processing 1,000,000 reviews monthly → cost per review ≈ ₹0.0023

Pricing Strategy

Startup Brands: ₹2,000/month

Growing Brands: ₹10,000/month

Enterprise Clients: ₹50,000/month

Competitive Pricing Analysis

Brandwatch Analytics Tools: ₹40,000+/month

Mention Monitoring Tools: ₹15,000+/month

Manual Review Analysis Teams: ₹20,000–₹50,000/month

InsightWear AI: Starting at ₹2,000/month

This pricing structure reduces analytics costs by approximately 40–75% compared to existing solutions.

Break-even Analysis

With operational costs around ₹7500/month and a startup plan priced at ₹2000/month, the system reaches break-even with approximately four startup customers.

ROI Example

Monthly orders: 5,000
Average order value: ₹2,000
Monthly revenue = ₹1,00,00,000
Return rate = 15% → 750 returns
Operational cost per return = ₹500
Total monthly return loss = ₹3,75,000
If InsightWear reduces returns by 10% → Savings ≈ ₹37,500/month
Subscription cost = ₹10,000/month
Net gain ≈ ₹27,500/month (~3.7x ROI)

Cost Optimization Strategies

- Use lightweight NLP models for lower inference cost
- Batch processing for large review datasets
- Containerized deployment using Docker

• Efficient caching and optimized database queries

4. Scalability Analysis

Prototype stage: 10,000 reviews/day

Growth stage: 100,000 reviews/day

Enterprise scale: 1,000,000+ reviews/day

Scaling Strategy

• Docker containers for horizontal scaling

• Distributed worker queues for batch processing

• Load-balanced API servers

• Database sharding for high-volume datasets

Economies of Scale

As review volume increases, compute utilization improves through batch inference and optimized pipelines. This allows the effective cost per review to decrease from ₹0.0023 to approximately ₹0.0018 at large enterprise scale.

Potential Bottlenecks and Mitigation

Database read/write limits at high review volumes

Mitigation: Redis caching and distributed storage

Traffic spikes during large sales events

Mitigation: Auto-scaling worker nodes and queue systems

5. Implementation Feasibility

Setup Complexity: 5/10

Integration complexity: Amazon SP-API (~2 days), Myntra seller API (~1 day), CSV upload supported.

Time to Deployment: 2–3 weeks for MVP

User Experience and Learning Curve

The dashboard is designed for non-technical users such as product managers and QA teams. The interface requires minimal training with an estimated learning curve of less than one hour.

Maintenance includes periodic model updates, infrastructure monitoring, and system performance tuning.

6. Risk and Mitigation

Risk: Model accuracy variations across clothing categories

Mitigation: Continuous training using expanded datasets

Risk: Slow enterprise adoption

Mitigation: Free trials and simple API integrations

Risk: Cloud infrastructure cost spikes

Mitigation: Billing alerts, usage monitoring, and autoscaling limits

Risk: High review spikes during seasonal sales

Mitigation: Queue-based processing and dynamic worker scaling

7. Ethical and Privacy Guardrails

Customer reviews are anonymized and no personally identifiable information (PII) is stored.

The system processes only publicly available review text and follows privacy-first data processing practices.

The platform uses controlled NLP pipelines for aspect extraction rather than open-ended generative responses, minimizing risks of prompt injection or jailbreak attacks.

The system also includes safeguards to detect and filter suspicious automated review manipulation.